



School of Mathematics

Programmes in the School of Mathematics

Second Examination

Programmes involving Mathematics

Second Examination

Linear Algebra

Level I 2LA 06 15552

May/June Examinations 2024–25

One Hour and Thirty Minutes

Full marks will be obtained with complete answers to BOTH questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

1. Let $(V, \langle \cdot, \cdot \rangle)$ denote an inner product space with $\dim V = n$. Let $\|\cdot\|$ denote the norm induced by the inner product $\langle \cdot, \cdot \rangle$.

(a) Show that for all $\mathbf{u}, \mathbf{v} \in V$ there holds

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

For which non-zero vectors $\mathbf{u}, \mathbf{v}$ does equality hold? [4]

- (b) Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ denote a set of non-zero vectors in V , orthogonal with respect to the inner product $\langle \cdot, \cdot \rangle$.

(i) Show that S is a linearly independent set if $k \leq n$.

(ii) Show that $\mathbf{0} \in S$ if $k = n + 1$. [6]

In the remainder of this question, we let $\mathcal{P}_m(\mathbb{R})$ denote the space of polynomials of degree m with real coefficients. We define the inner product $\langle \cdot, \cdot \rangle : \mathcal{P}_4(\mathbb{R}) \times \mathcal{P}_4(\mathbb{R})$ via

$$\langle p, q \rangle := \int_{-1}^1 p'(x)q'(x)dx + p(0)q(0),$$

where $p'(x), q'(x)$ denote the first derivatives of $p(x), q(x)$, respectively, with respect to x .

- (c) Let $V = \mathcal{P}_3(\mathbb{R})$. Find a basis $B = \{q_1, q_2, q_3, q_4\}$ for V which is orthogonal with respect to the inner product $\langle \cdot, \cdot \rangle$. [8]

- (d) Find a polynomial q_5 orthogonal to the polynomials q_i ($i = 1, 2, 3, 4$) from part (c). Make sure you explain your working or justify your answer. [3]

- (e) Find the orthogonal projection of $p(x) = x^4$ onto V with respect to $\langle \cdot, \cdot \rangle$. [4]

2. Let $f : V \rightarrow V$ be a linear map, where V denotes a finite-dimensional vector space over some field $\mathbb{F}$.

(a) Let $a \in \mathbb{F}$ and define $E_a := \ker(a \cdot \text{id}_V - f)$, where id_V is the identity map on V .

(i) Show that if a non-zero vector $\mathbf{v}$ is in E_a for some a , then $(a, \mathbf{v})$ is an eigenpair for f .

(ii) Show that $E_a \leq V$.

(iii) Can we have $E_a = V$? If so, give an example; if not, explain why. [8]

Let now $V = \mathcal{P}_2(\mathbb{R})$ and let f be given by

$$f(p) = (x-1)p'(x), \quad p \in V.$$

(b) Show that f is a linear map. [2]

(c) Write down the matrix representation A for f with respect to the power basis. [3]

(d) Write down the characteristic polynomial of A . Hence, find the eigenpairs of A . [9]

(e) Use the eigenvectors of A to derive the eigenvectors of f . [3]

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Linear Algebra

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g., notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g., in your pockets or in your pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part; if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules, may be subject to the Student Conduct procedures.